



Amey Kemkar
Bachelor of Technology
in Mechanical Engineering
Indian Institute Of Technology, Ropar

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GitHub
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EDUCATION

Degree	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	Indian Institute of Technology, Ropar	8.79	2024-2028
Senior Secondary	Central Board of Secondary Education	94.2%	2024
Secondary	Indian Council for Secondary Education	98.5%	2022

PROFESSIONAL EXPERIENCE

- Research Internship at Applied Plasma and Space Propulsion Lab** June 2026 - July 2026
under Dr. Pravendra Kumar, Department of Aerospace Engineering, IIT Madras.
 - Built a four-week foundation in **plasma physics** for **RF ion thrusters**, covering single-particle motion, fluid and kinetic models, wave-particle interactions, and the Vlasov equation.
 - Explored the **numerical methods** deployed to simulate plasma under various conditions and to understand the results obtained from these simulations for **post-processing** and **optimization**.
 - Gained exposure to **experimental propulsion research** through **electrospray experiments** at APSPL and for **Combustion** via visits at the National Centre for Combustion Research and Development (NCCRD).

PROJECTS

- Particle-in-Cell Simulation of RF Ion Thruster Plasma Dynamics** Technical Report | Github
at Applied Plasma and Space Propulsion Lab, under Dr. Pravendra Kumar June 2026 - Present
 - Built a 1-D Electrostatic and Electromagnetic (**ES-1 and EM-1**) **PIC Solver** in Python based on Boris Pusher, FFT Poisson solver and YEE FDTD algorithms.
 - Simulated plasma oscillations, upper hybrid resonance, and Landau damping within 5% of theory; identified timestep limits on cyclotron gyration via error analysis.
 - Benchmarked **PICFoam** as a path to scalable multi-dimensional RF discharge modeling beyond the 1D solver.
- Design of a Powered-Lift Wing & Propeller System** Technical Report | Github
LAT Aerospace- High Prep Problem Statement, Inter-IIT 14.0 Sept 2025 - Dec 2025
 - Extended thin-airfoil vortex-sheet theory to cambered blown-flap airfoils with jet interaction, embedded in a Prandtl lifting-line framework for finite wing analysis.
 - Optimized flap chord ratio, deflection angle, and angle of attack via constrained optimization, achieving **CL = 7.06** and **L/D = 5.10** at takeoff and **L/D = 57.97** at cruise, exceeding both target design constraints.
 - Cross-validated** theoretical lift predictions against **OpenVSP** panel method results before CFD handoff.
- Deep Learning-based Structural Stress Prediction and Topology Optimization** Github
under Dr. Manish Agrawal May 2026 - June 2026
 - Trained MLP and CNN surrogate models to **predict peak von Mises stress** on lattice cantilever structures (best **R² = 0.76**), as a fast alternative to repeated FEA.
 - Used backpropagation-derived stress sensitivities with gradient descent (**R² = 0.62**) and a novel spatial gradient descent (**R² = 0.67**) to optimize material layouts for minimum peak stress.
 - Designed a **U-Net** for full-field von Mises stress prediction and benchmarked it against direct CNN regression.

TECHNICAL SKILLS

- Programming Languages and Libraries:** C++, Python, NumPy, Matplotlib, Tensorflow, Scikit-Learn
- Simulation and Modelling Softwares:** MATLAB, Simulink, SolidWorks, ANSYS Mechanical, ANSYS Fluent, OpenFOAM, PICFoam, XFOIL, OpenVSP, VSPAERO, OpenRocket, OpenMotor
- Engineering Areas:** Electric Propulsion & Plasma Simulation, CFD, Aerodynamics, Rocket Propulsion, CAD Design, Structural Analysis, Machine Learning, Deep Learning

KEY COURSES TAKEN

- Core Mechanical:** Solid Mechanics, Thermodynamics, Machine Drawing, Fluid Mechanics, Theory of Machines, Deep Learning in Physical Systems
- Mathematics and Miscellaneous:** Probability and Statistics, Linear Algebra, Differential Equations, Calculus, Introduction to Computing and Data Structures, Basic Electronics

POSITIONS OF RESPONSIBILITY

- Representative,** Zenith-The Astronomy and Space Tech Club, IIT Ropar Apr. 2025 - May, 2026
- Mentor,** Institute Student Mentorship Program, IIT Ropar May, 2025 - May, 2026

MISCELLANEOUS

- Top 7% Merit Scholarship,** Awarded for academic performance, ranking among the top 7 % at IIT Ropar. 2025
- Contingent Member,** Represented College at Inter-IIT Tech. Meet 14.0 2025
- Trinity Grade-4 Electronic Keyboard,** Passed the Grade Exam Level 4 in Electronic Keyboard with Distinction 2019